

TEXT — WHAT YOU ALREADY DO

MORE COMPLEX

What You Can Make with Desktop AI Apps

faculty projects, ordered from familiar text toward unfamiliar code — and past it

A review of what faculty are building, arranged on one axis. It begins in the writing you already command and descends toward forms that did not exist two years ago. Each row is a genre of output. Examples at left. One project **unpacked** at right: situation, inputs, operations, outputs, and the interface suited to it ([CHAT](#) [DESKTOP](#) [CODE](#)). The loop is constant — inputs, operations, outputs. Only the medium changes. Begin where you are fluent. Descend.

1 Structured Text & Documents

EXAMPLES

- Email to a chair proposing a new course
- Grant Specific Aims page from a project description
- Committee report from raw meeting notes
- Recommendation letter from a CV and a prompt
- Conference abstract from a draft paper
- Syllabus skeleton from a reading list

UNPACKED

Meeting notes → a clean report for the dean

SITUATION	Messy notes from a faculty meeting; a structured report due tomorrow.
INPUTS	meeting-notes.md — your raw bullet points.
OPERATIONS	One prompt: "turn these into a report with a summary, decisions made, and action items with owners and dates."
OUTPUTS	report.md , or a polished .docx on request.
BEST FIT	CHAT DESKTOP CODE IF REPEATED MONTHLY

2 Reference Pages

static HTML

EXAMPLES

- A searchable glossary of course terms
- A one-page syllabus website
- A printable command/reference dashboard
- An illustrated lecture handout
- A course-policy FAQ page

UNPACKED

A glossary your students can search and copy from

SITUATION	One page holding every key term in your course, searchable, works offline.
INPUTS	glossary.md — terms and definitions.
OPERATIONS	"Build a single-file HTML page with a search box and click-to-copy, styled for comfortable reading."
OUTPUTS	glossary.html — one file you can email or host anywhere.
BEST FIT	CODE DESKTOP

3 Web Apps

Next.js

EXAMPLES

- A course website with interactive concept demos
- A reading-response collection app
- An interactive timeline of a period or movement
- A "build your own X" student exercise

UNPACKED

Course site for GENED 1049 East Asian Cinema

SITUATION	A site pairing a cinematography glossary with demos built on the films you actually teach.
INPUTS	Syllabus, <i>Rashomon</i> stills, glossary text.
OPERATIONS	Scaffold a Next.js app, build parametric demo components, deploy to Vercel.
OUTPUTS	A live site (gened-1049.vercel.app) you share by link.
BEST FIT	CODE

Turn over. The axis continues: connected tools, visualization, orchestration. Then it leaves the page.

...and further down the spine

connected tools, visualizations, agentic systems — and what's past the bottom edge

4

Bots & Connected Tools

MCP, Slack

NOT COVERED TODAY

EXAMPLES

- An assistant connected to a library catalogue for live literature search
- An art-history lecture pulling live images from a museum's own records
- A Slack bot that answers from your course materials
- An oral-exam practice bot
- An assistant connected to your Google Drive

WORTH KNOWING

AI assistants can be wired to live services

Every row above works on files you already have. This one is different: the assistant reaches *out* — to a catalogue, a database, an institutional system — and works with whatever it finds there. The plumbing may use **MCP** or a provider-specific connector, and access has to be deliberately set up and granted before the assistant can use it.

We are not setting one up today. Worth knowing the category exists — and worth knowing that connecting an AI assistant to a live service is a real grant of access to that account, not a setting you flip casually.

5

Data Visualizations

d3, three.js

EVEN MORE COMPLEX

EXAMPLES

- A 2D semantic-embedding map of a text corpus
- An interactive chart of course-evaluation trends
- A 3D model of a molecule, artifact, or building
- A network graph of citations or characters
- An animated explainer for a hard concept

UNPACKED

Calvino's *Six Memos*, by the numbers

SITUATION	Students see literary qualities as a measurable space — and plot their own prose against it.
INPUTS	The five memos as text; definitions for each measure.
OPERATIONS	Deterministic text analysis + a 2D embedding map + a live draft composer, charts drawn in d3.
OUTPUTS	A live page where pasted prose appears in every chart in real time.
BEST FIT	CODE

6

Agentic Pipelines & Systems

MOST COMPLEX

EXAMPLES

- A make-up exam generator that interviews you between drafts
- A reproducible coding pipeline for interview transcripts
- Close-reading a whole corpus with parallel subagents
- An automated weekly research digest (scheduled)

UNPACKED

A make-up exam that is genuinely equivalent, not a reshuffle

SITUATION	A student misses the final. You need a replacement that tests the same things at the same difficulty, and you do not have a week.
INPUTS	The original exam, as LaTeX and PDF.
OPERATIONS	A skill runs a multi-round flow: read the exam, interview you on what each question is really testing, draft two or three candidates per slot, revise on your feedback, then assemble the chosen ones in the original's format.
OUTPUTS	A candidate bank you chose from, plus the finished make-up exam as LaTeX.
BEST FIT	CODE MULTI-ROUND, STATEFUL

? Unknown ???

BEYOND THE MAP

One day, a sort of Gesamtkunstwerk. Past the bottom: potentially new mediums, or new combinations of the rows above — text, application, visualization, sound, physical space in one project — buildable as the remaining technical barriers come down. Too early to catalogue.

The whole axis is one loop: inputs, operations, outputs. Descent changes a single variable — how much of the loop is code. Worked versions of most rows live in this repo's **02-further-use-cases/**.